

CHAPTER 3

SITE INVESTIGATION

3.1. General

Many projects exceed their budgets and their completion dates due to unforeseen problems during the excavation and construction of their foundations. To ensure that these problems are kept to a minimum, a thorough site investigation is required, particularly for tall buildings. The collapse of the deep excavation for the Circle Line at Nicoll Highway and Fusionpolis in 2004 sent reminders on the importance of a thorough site investigations and initiated an urgent and stringent review on the adequacy of existing regulations governing the design and construction of major geotechnical works especially temporary earth retaining structures. A site investigation is a study of the environment and the ground conditions required for any engineering or building structure. It is a process by which geological, geotechnical, topographical, social, environmental, economic and other relevant information are collated and analysed. The investigation may range in scope from the study of maps or aerial photographs, site reconnaissance, a simple examination of the surface soils with or without a few shallow trial pits, to a detailed study of the soil and groundwater conditions to a considerable depth below the surface by means of boreholes and *in situ*/laboratory tests on the materials encountered.

The extent of the work depends on the importance and foundation arrangement of the structure, the complexity of the soil conditions, and the information which may be available on the behaviour of existing foundations on similar soils. Thus it is not a normal practice to sink boreholes and carry out soil tests for single or two storey dwelling houses or similar structures since there is usually adequate knowledge of the required foundation depths and bearing pressures in any particular locality.

Sufficient information to check the presumed soil conditions can usually be obtained by examining open sewer trenches or shallow excavations for roadworks, or from a few shallow trial pits or hand auger borings.

A detailed site investigation involving deep boreholes and laboratory testing of soils is always a necessity for heavy structures such as bridges, multi-storey buildings or industrial plants [1,2].

3.2. Information to be Retrieved

In a fairly detailed study, the following information should be obtained in the course of a site investigation:

- (a) The general topography of the site as it affects foundation design and construction, e.g. surface configuration, adjacent property, the presence of watercourses, ponds, hedges, trees, rock outcrops, etc., and the available access for construction vehicles and plants.
- (b) The location of buried services such as electric power and telephone cables, water mains and sewers.
- (c) The general geology of the area with particular reference to the main geological formations underlying the site and the possibility of subsidence from mineral extraction or other causes.
- (d) The previous history and use of the site including information on any defects or failures of existing or former buildings attributable to foundation conditions.
- (e) Any special features such as the possibility of flooding, seasoning swelling and shrinkage, soil erosion, etc.
- (f) A detailed record of the soil and rock strata and ground conditions within the zones affected by foundation bearing pressures and construction operations.
- (g) Results of laboratory tests on soil and rock samples appropriate to the particular foundation design or constructional problems.

3.3. Stages of Site Investigation

The stages of a site investigation are similar for most projects although the content and number generally increase with the size of the project and the complexity of the ground conditions. Figure 3.1 shows the general

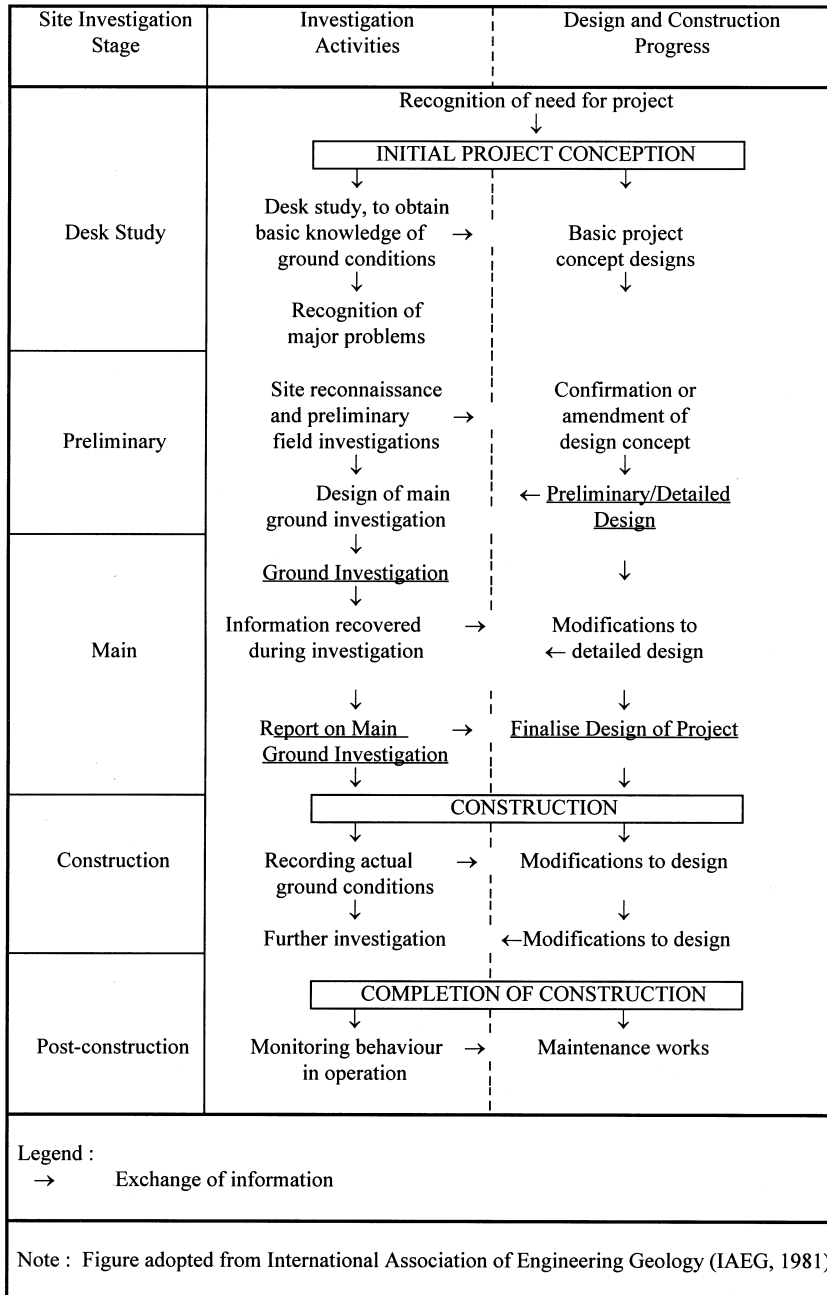


Figure 3.1. The investigation sequence of events.

investigation sequence of events from the inception to the completion of a project [3].

For a given site, some of the stages may be unnecessary, overlap or be taken out of the sequence. For example, site reconnaissance may take place before the completion of the desk study. There are generally four stages of site investigation, namely:

- (a) Planning
- (b) Desk Study
- (c) Site Reconnaissance
- (d) Ground Investigation

3.4. Planning

Planning is needed for all stages of a site investigation. The client's brief and the design engineer's design thoughts should be coordinated before the desk study. Knowledge of the design proposals should include layout, alignments, function and probable loadings. Once these points have been established by discussion with the design engineer, the geotechnical or site investigation contractor can proceed to gather preliminary information.

3.5. Desk Study

The desk study involves the collection of available documentary materials relevant to the site, the immediate environment and the proposed structure in any of the following forms:

- (a) Previous Ground Investigation
 - Useful information may be extracted which will greatly assist the planning of the ground investigation, and may reduce the scope and extent of the investigation required. It should be noted, however, when interpreting these data, that the methodology, procedure and requirement at that time may be different.
- (b) Topographical Maps
 - This is useful for undeveloped areas such as forests and swampy or mountainous areas where information of the following general features can be obtained:

- Past uses of site.
- The location of roads, railways, tracks, rivers, canals, foot-path, pipelines, overhead lines, contour lines, etc.

(c) Aerial Photos

- Aerial photos can provide a detailed and definitive picture of the topography, lines of communication (roads, railways and canals), surface drainage and urban development.

(d) Geological Maps and Memoirs

- These provide detailed information of the geology of the district, and are useful as a basis for evaluating the likely influence of the local geology on the proposed works and in the selection of the ground investigation methods. These are used frequently in Singapore and such maps are provided in a publication by the then Public Works Department (PWD) titled “Geology of the Republic of Singapore” [4,5].

(e) Historical Maps

- These provide information on the past uses of the site which could lead to identification of buried river courses or previous swampy ground. These would be important in delineating the possible problem areas which will need more detailed investigation during the main ground investigation. As a result of the extensive land reclamation and associated removal of high ground in Singapore over the decades, the present topography is frequently unrepresentative of the actual geological formations. Some examples of the historical maps from the National Archives are listed in Table 3.1.

(f) Rainfall Records

- This is particularly important for works involving basement construction, slope and major drainage system. Hydrological information is useful in drainage studies, including the assessment of flooding risk and the influence of proposed works on the local and downstream drainage regimes. Climatology information can be obtained from the Meteorological Services Division, National Environmental Agency.

Table 3.1. List of some Historical Maps available in National Archives, Singapore.

Reference No.	Publication Date	Description
216	1836	Map of Town & Environment of Singapore by G. D. Coleman
99	1854	Old map of Singapore
2152	1845	Southern Part of Singapore
4	1865	Map of Island of Singapore & its dependencies drawn by J. Van Cuylenburg, Surveyor General Office, Singapore
217	1873	Map of Island of Singapore & its dependencies showing various administration districts
14	1873	Map of Island of Singapore & its dependencies drawn by Wajud Khan, Surveyor General Office, Singapore
106	1886	Map of Singapore & its dependencies drawn by Wajud Khan
13	1858	Map showing Government House
12	1905	Map showing Government House
6	1506	Map of Singapore Town with Municipal Limits
15	1526	Singapore Road Map — FMS Surveys No. 34
11	1527	Singapore published under direction of Surveyor General FMS & SS
16	1532	Map of Singapore Town FMS Survey No. 15, 1532
415	1532	Singapore showing City area
416	1532	Singapore showing Kallang & Paya Lebar
405	1543	Map of Singapore Town — Tiong Bahru area
140	1951	Singapore by Survey Department Singapore
420	1957	Map of Singapore, Survey Department Federation of Malaya
17	1961	Singapore printed by the 84 Survey Squadron RE Far East

(g) Geographic Information System (GIS)

- The most important GIS at the national level is the Singapore Land Data Hub project administered by the Singapore Land Authority (SLA), Ministry of Law [6,7]. The project was conceived in 1989 as a multi-ministry effort to establish a central repository of accurate and comprehensive digitised

land data. The digitised land data in the hub are contributed by more than 15 public sector agencies. These agencies have the relevant business processes within their operations to keep the digitised land data updated. The digitised land data include the survey maps (the legal land boundary); buildings and roads information; infrastructure data such as drainage and sewerage; utilities information such as electricity, gas and water network; electronic street directory data etc. [7]. SLA hosts an online service — Integrated Land Information Service (INLIS) which provides information including survey plans, road line plans, surrounding amenities, horizontal and vertical control points.

3.6. Site Reconnaissance

This is to confirm, amplify and supplement the information collected earlier by (a) having a visual investigation on the site, and (b) collecting information from local inhabitants.

In the visual investigation, the following features are observed:

Topography: The general topography of the site is observed for evidence of the soils present, their distribution and their properties. Instability such as soil creep especially on sloping sites must be noted. Information on undulations, tilted tress, evidence of movement of buildings or deformation should be gathered.

Construction Materials and Labour: Especially for large countries, the cost of transporting bulk construction materials such as sand and aggregates as well as labourers can be excessive if there is no nearby source available. Accommodation, transport, food, amenities and facilities will have to be provided.

Groundwater Conditions: The presence of rivers, canals, springs and seepages, areas of wet ground, shallow wells, vegetational features can provide information on the water table conditions.

Site Access: This is to confirm the problems with access during the desk study and solutions formulated.

Surrounding Structures: In cases where the neighbouring lands have been developed, the sensitivity of the proposed project affecting the existing structures must be evaluated. Regulatory restrictions on

movement/vibration and noise allowable from a construction site in a way may determine the types of foundation works to be employed later. Minimum vibration is allowed, for instance, if there is an underground tunnel nearby (Building Works and Restricted Activities in Railway Protection Zone [8–10]). Minimum noise is allowed if there is a hospital, residential area nearby (Environment Public Health Act [11, 12]).

Useful information can also be collected from local authorities, local statutory boards and government bodies, local archives, local inhabitants, local contractors, local clubs and societies, schools, colleges and universities.

3.7. Ground Investigation

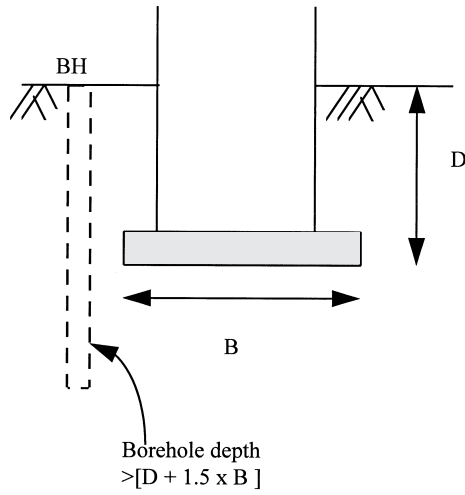
Ground investigation is to determine (a) the suitability of the site for the proposed project, (b) an adequate and economic foundation design, (c) the difficulties which may arise during construction and (d) changes and cause of changes in subsoil conditions.

Geotechnical engineers specialising in soil sampling and testing are engaged to establish the parameters that will be used in the design of the building foundation. The amount of testing depends on the size and complexity of the structure, the type of soil encountered, proximity of the proposed structure to existing buildings, and the level of the groundwater table.

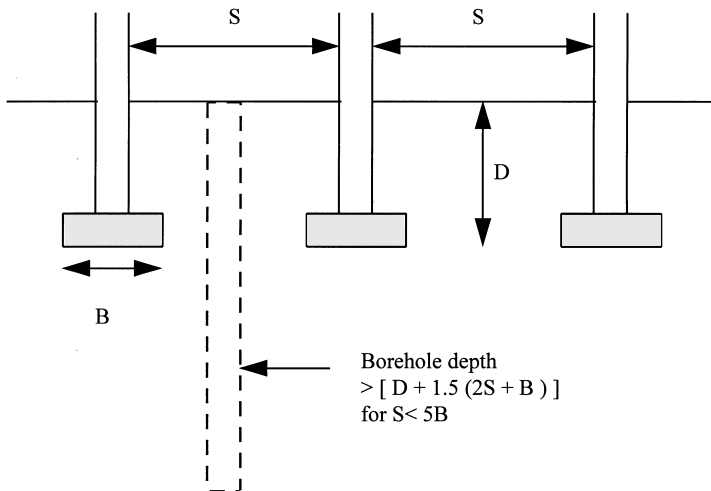
Other considerations are the scope of the proposed work, the amount of existing information available, the probable nature and variability of ground conditions, the availability of plant and equipment, the cost of investigation, the manpower for operation and supervision, access limitations, and temporary works required.

The degree to which the proposed development affects the ground condition may provide a guide to the scope of investigation (Figure 3.2). In the case of a large structure where many piles are clustered together, the much larger combined pressure bulb indicates that a much more detailed investigation is needed [13–16].

Investigative methods range from non-intrusive (non-destructive) geophysics through to intrusive borehole installations. The methods can be broadly divided into (a) field tests and (b) laboratory tests.

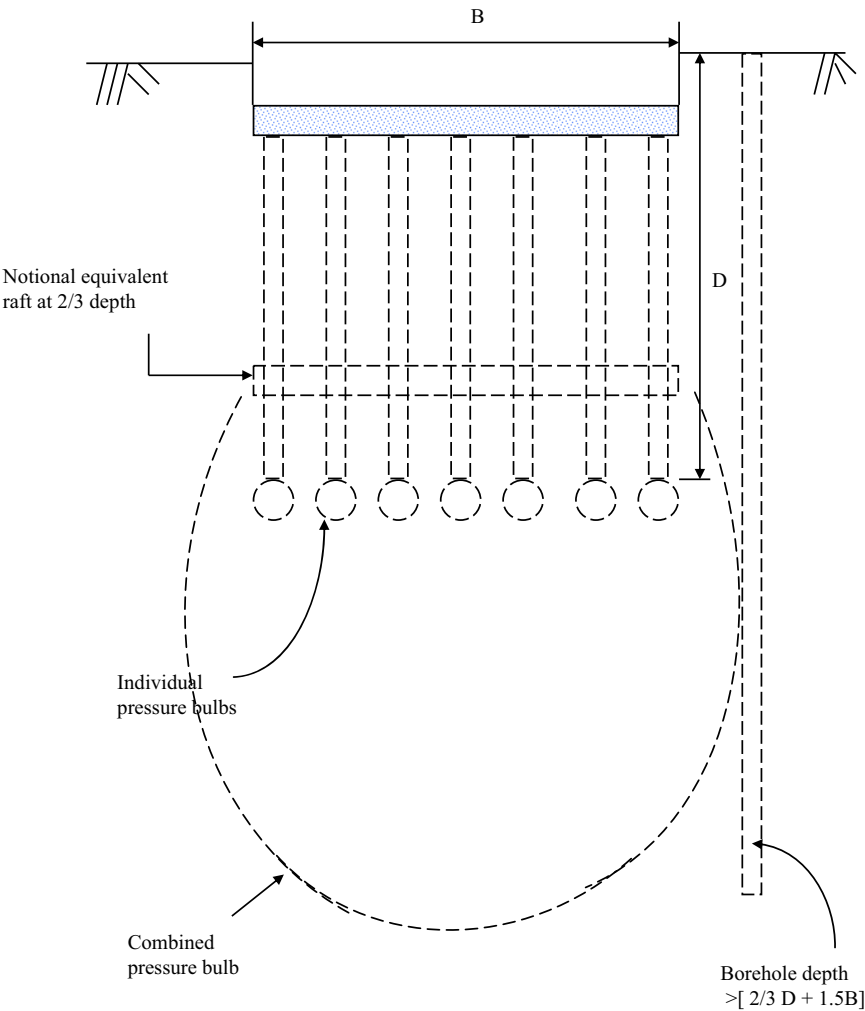


(a) Structure on isolated pad or raft



(b) Closely spaced strip on pad footings

Figure 3.2. Estimated borehole depths based on pressure bulb [14].



(c) Large structure on friction piles

Figure 3.2. (Continued).

3.7.1. Field Tests

The common on-site ground investigation techniques include the following:

3.7.1.1. Subsurface Mapping

A Ground Penetrating Radar (GPR) (Figure 3.3) can be used to explore the subsurface of the ground non-destructively to map geologic conditions e.g. depth to bedrock, depth to water table [17], depth and thickness of soil and sediment strata on land and under fresh water bodies [18, 19], the location of subsurface cavities and fractures in bedrock [20]. It can also be used to locate objects e.g. electrical conduit, water and sewer lines.

GPR works by transmitting pulses of ultra-high-frequency electromagnetic (typically from 10 MHz to 1,000 MHz) waves into the ground through a transducer or antenna. The transmitted energy is reflected from buried objects, captured by a receiving antenna and stored in a digital control unit for later interpretation.



Figure 3.3. A ground penetrating radar (GPR).

3.7.1.2. Trial Pitting

Trial pits are used where only shallow depths are to be investigated. Trial pits up to 5 m can be quickly and cheaply dug using a hydraulic backactor. It allows a visual inspection of *in situ* soil conditions both laterally and vertically and allows a detailed examination of soil variability, structure and weathering profile.

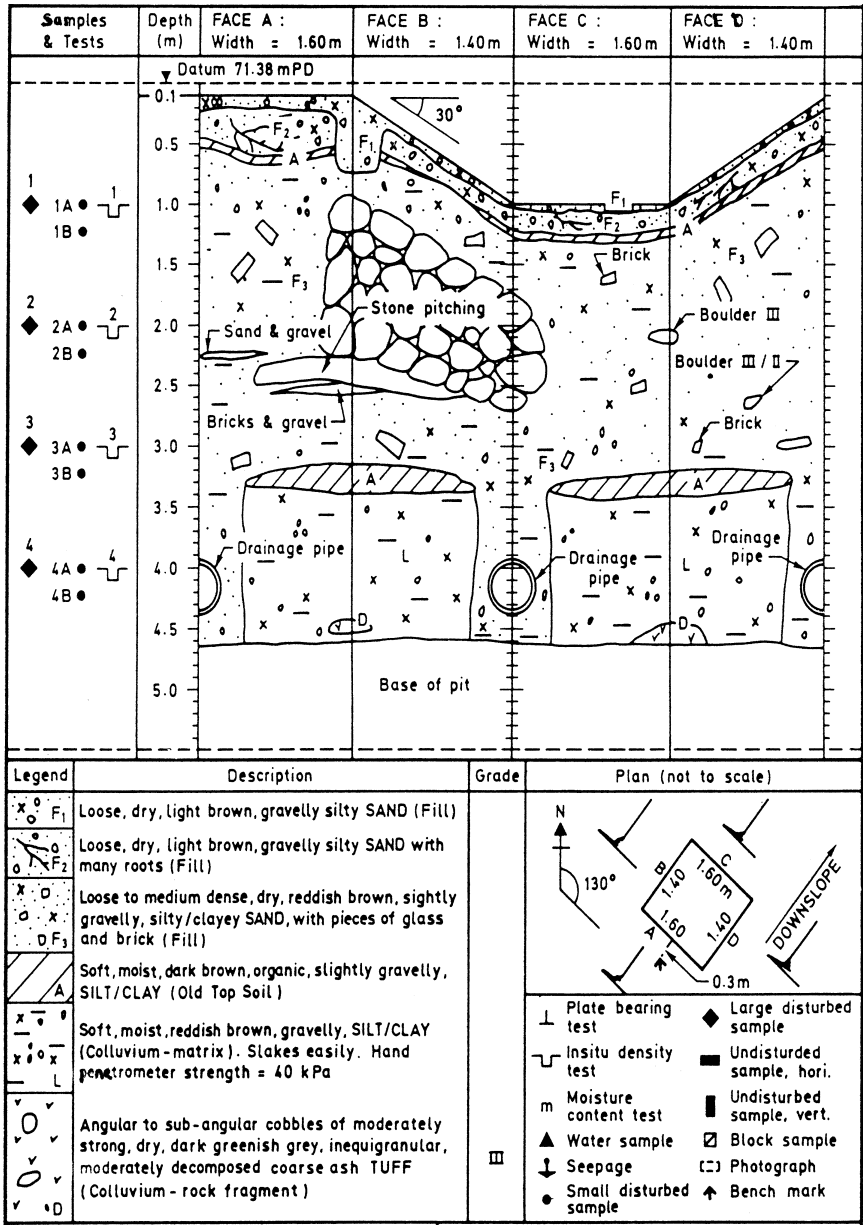


Figure 3.4. Example of a trial pit log.

The field record of a trial pit should include a plan giving the location and orientation of the pit, and a dimensioned section showing the sides and the floor. Figure 3.4 shows an example of a trial pit log.

The limitations of trial pitting are obvious:

- Excavation below water table is difficult
- Not applicable for certain soil such as hard rock
- May cause some areas of ground disturbance
- Supports are needed for deep trial pits such as trenches

3.7.1.3. *Exploratory Boreholes*

Boreholes or deep boring, are sunk to collect soil samples in the case where investigation requires deeper soil samples than can be excavated by trial pitting. Boreholes can be formed by hand augering, light cable percussion boring, mechanical augering, rotary open hole and rotary core drilling [21–25]. The details of boring, sampling and testing are described in BS5930:1999 [26]. A borehole usually involves works including boring through soil, coring through boulders, sampling, dynamic probe testing, in situ testing, and water-table observations. Figure 3.5 shows an example of a boring log. Figure 3.6 shows a typical cross section of soil profile of a site, derived from information retrieved from various boreholes.

3.7.1.4. *In Situ Testing*

Mackintosh probe is often use in preliminary site investigation. It is a quick and cheap method to acquire an indication of the depth to a hard stratum. It consists of a steel rod with a bullet shaped tip, hammered into the ground using a constant weight (Figure 3.7(a)). The number of blows required to drive the probe to a predetermined depth is referred as the N value. A variety of small samplers, augers and probe heads can be attached to the probe for sampling purposes.

In situ Standard Penetrating Test (SPT) is commonly used in boreholes to obtain the relative density and consistency of the soil encountered at different levels. As per BS1377 [27], a hammer weight of 65 kg having a drop of 760 mm is driven to a total penetration of 450 mm into the soil and the number of blows for the last 300mm is taken as a measure of the soil resistance SPT 'N' value.

LOG OF BORING

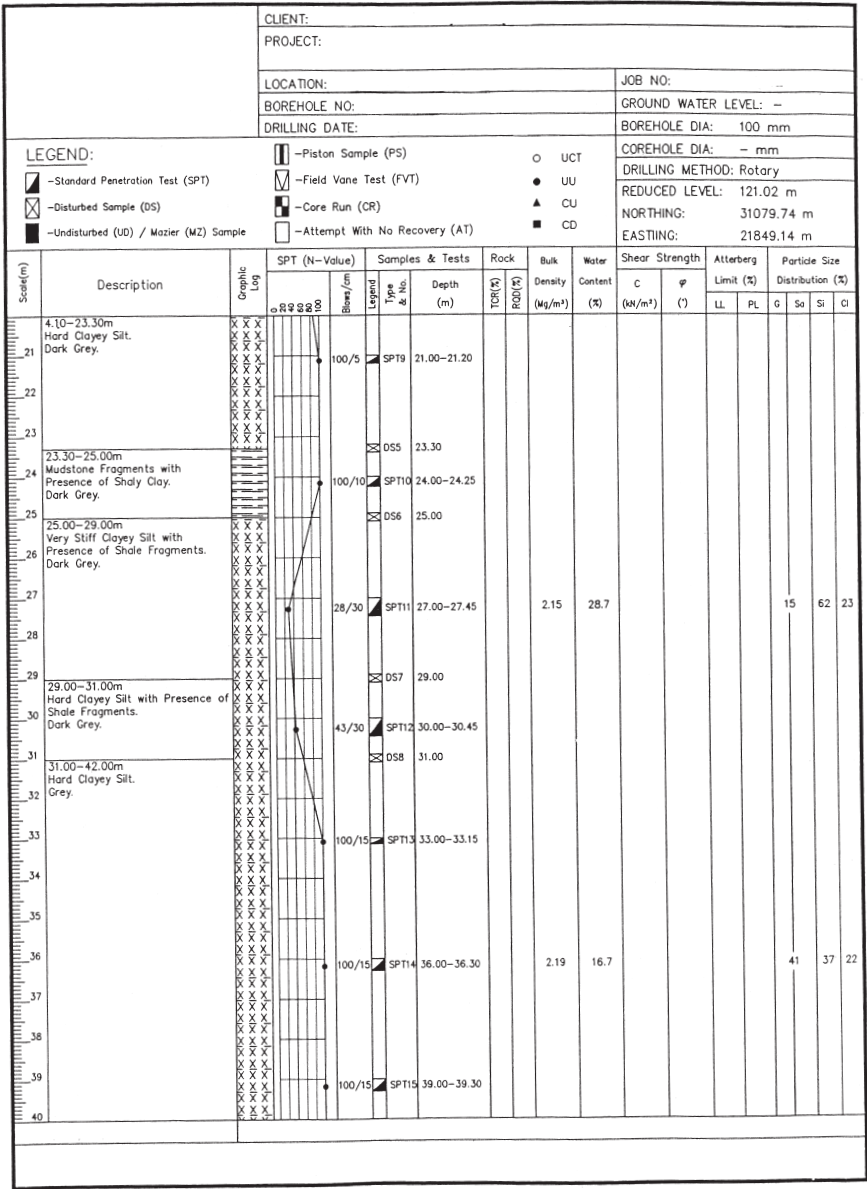


Figure 3.5. Example of a boring log.

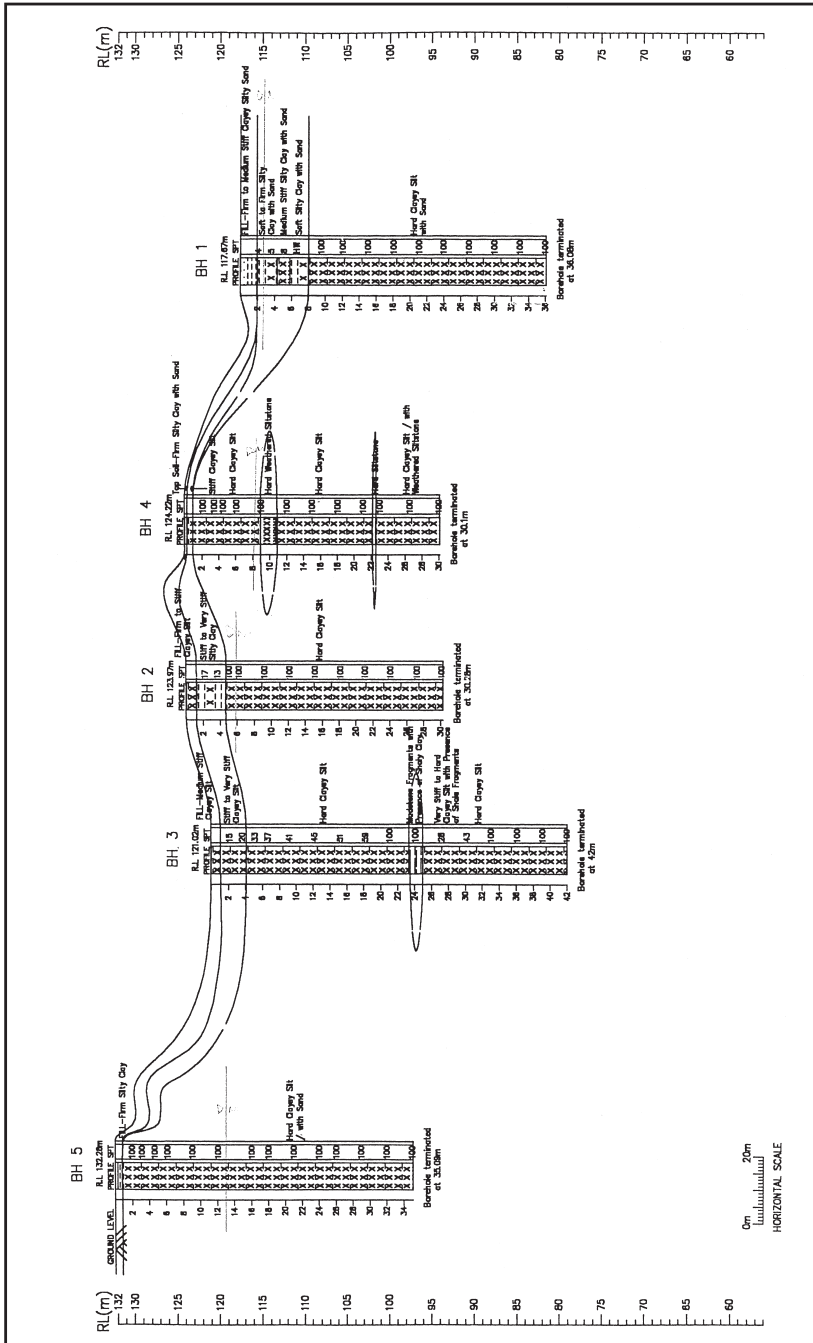


Figure 3.6. Cross section of soil profile through various boreholes.



Figure 3.7(a). Mackintosh probes test.



Figure 3.7(b). California Bearing Ratio (CBR) laboratory tester (courtesy: ELE International).

A cone penetrometer can be used to estimate the California Bearing Ratio (CBR). CBR is a simple strength test that compares the bearing capacity of a material with that of a well-graded crushed stone (Table 3.2 and Figure 3.7(b)). A small cone on the end of a rod is pushed steadily into the ground at a constant rate of penetration and the approximate CBR read off a moving scale at the top [28].

Plate bearing test can be carried out at ground level or in a trial pit. The test involves loading a plate and measuring its penetration into the ground thus giving the stress–strain relationship for the soil (see Figure 3.8). A loaded lorry may be used as the fixed support for the hydraulic jack.



Figure 3.8(a). Plate bearing test (courtesy: ELE International).



Figure 3.8(b). Plate bearing test (courtesy: Wiratman & Associates).

A Vane tester (Figure 3.9) can be used to estimate the shear strength of clayish soils. The tester measures the torque required to shear a small cylinder of soil within the soil mass.



Figure 3.9. A Field Vane Shear tester (courtesy: ELE International).



Figure 3.10. A ring penetrometer (courtesy: ELE International).

A ring penetrometer is another easy method to give an estimation of the shear strength of clayish soils. The plunger is gently pushed in as far as the cut graduation line and the maximum reading read off a sliding indicator on a marked scale (Figure 3.10).

3.7.2. Laboratory Tests

Laboratory tests for the collected samples are required to identify and classify the samples in terms of type, age, composition, weathering and moisture content, and to make an evaluation on the soil behaviour [29–35].

An example of a laboratory summary report is shown in Table 3.2. The various tests categories and descriptions are summarised in Table 3.3 [36–40].

3.8. Site Works and Setting Out

After a site is handed over, the task of (a) clearing the site, (b) setting out the building and (c) establishing a datum level can commence. A land

Table 3.2. An example of a laboratory summary report.

Borehole No.	Sample No.	Depth (m)	Bulk Density (Mg/m³)	Dry Density (Mg/m³)	Moisture Content (%)	Vane		U.C.T	Triaxial Compression			Consolidation			Atterberg Limits			Particle Size Distribution				Particle Density (Mg/m³)	Chemical								
						Undisturbed Shear Strength (kN/m²)	Remoulded Shear Strength (kN/m²)		U.U	C.U	C.D	Cohesion (kN/m²)	Phi Angle (°)	Cohesion (kN/m²)	Phi Angle (°)	Initial Void Ratio	Degree of Saturation (%)	Preconsolidation Pressure (kN/m²)	Liquid Limit (%)	Plastic Limit (%)	Plasticity Index (%)		Gravel (%)	Sand (%)	Silt (%)	Clay (%)	Total Sulphate Contents (%)	Organic Matter Content (%)	Chloride Content (%)	pH Value	Sulphate Content in 2:1 water:soil extract
BH 1	UD1	1.50-1.95	2.07	1.77	17.2														3	47	50										
	UD2	3.00-3.45	2.05	1.64	25.0														6	48	10	36									
	UD3	4.50-4.95	2.00	1.56	27.9				43	0																					
	UD4	6.00-6.45	2.06	1.70	21.3				18	0																					
	UD5	9.00-9.45	2.10	1.79	17.0				117	0																					
	SS3	4.95-5.40	-	-	-																										
	SS4	6.45-6.90	2.17	1.89	15.0																										
	SS6	12.00-12.28	2.06	1.91	8.0																										
	SS7	15.00-15.25	-	-	-																										
	SS10	24.00-24.10	2.08	1.89	9.8																										
	SS13	33.00-33.13	2.26	2.03	11.6																										
	BH2	UD1	1.50-1.95	1.95	1.46	33.4				100	0										25	21	54								
UD2		3.00-3.45	2.00	1.59	25.8				60	0										1	21	23	55	2.69							
UD3		4.50-4.80	2.00	1.60	25.0						10	24								0	24	40	36								
SS2		3.45-3.90	-	-	-																										
SS4		6.00-6.45	2.13	1.89	12.8															31	36	33	2.68								
SS7		15.00-15.38	-	-	-																										
SS8		18.00-18.35	2.11	1.86	13.5															49	23	28									
SS11		27.00-27.42	2.18	1.84	18.3															37	39	24									

Client :
Location :
Job No. :

* Multi Stage
^ Field Vane
~ Lab. Vane
< Less Than

U.C.T – Unconfined Compression Test
U.U – Unconsolidated Undrained Triaxial Compression Test
C.U – Consolidated Undrained Triaxial Compression Test With Pore Pressure Measurement
C.D – Consolidated Undrained Triaxial Compression Test With Volume Change Measurement

Table 3.3. Tests on soils and groundwater.

Category of Test	Name of Test	Remarks
Soil Classification Tests	Moisture content	Frequently used in the determination of soil properties, e.g. dry density, degree of saturation. Soils containing holloysitic clays, gypsum or calcite can lose water of crystallisation when heated, and should be dried at various temperatures to assess the effect on determination of moisture control.
	Liquid and plastic limits (Atterberg limits)	Used to classify fine-grained soil and as an aid in classifying the fine fraction of mixed soils. Soils containing holloysitic clays must be tested at natural moisture content.
	Linear shrinkage	Used to detect the presence of expansive clay minerals.
	Specific gravity	Frequently used in the determination of other properties, e.g. void ratio, particle size distribution by sedimentation.
	Particle size distribution: (a) Sieving	(a) The common tests used are Sieve Analysis for content of sand and gravel (BS 1377, 1990 [27]), and Hydrometer Test for content of silt and clay. Core is required with soils derived from <i>in situ</i> rock weathering, to avoid crushing of soils grains during disaggregation.
	(b) Sedimentation	(b) The proportion of the soil passing the finest sieve (64 µm) represents the combined silt and clay fraction. The relative proportions of silt and clay can only be determined by sedimentation.
Soil Permeability Tests	Laboratory vane shear	A useful test for classifying silts and clays in term of consistency (Figure 3.11).
	Permeability: (a) Constant head permeability test (b) Falling head permeability test (c) Flexible wall permeability test (triaxial test) (d) grain size analysis (sieve analysis)	The permeability test is a measure of the rate of flow of water through soil. In a constant head permeability test, water is forced by a known constant pressure through a soil specimen of known dimensions and the rate of flow is determined. In a falling head permeability test, water is forced by a falling head pressure instead [41]. The constant head test is suited only to soils of permeability roughly within the range 10^{-4} m/s to 10^{-2} m/s. For soils of lower permeability the falling head test is applicable.

Table 3.3. (Continued).

Category of Test	Name of Test	Remarks
Soil Compaction Tests	Dry density/moisture content relationship	Indicates the degree of compaction that can be achieved at different moisture contents and with different compactive effort.
Chemical and Corrosivity on Soils and Groundwater	Organic matter content	Detects the presence of organic matter, which can: (i) interfere with the hydration of Portland cement in soil/cement pastes. (ii) influence shear strength, bearing capacity and compressibility. (iii) influence the magnitude of the correction factor required when using nuclear methods to estimate the <i>in situ</i> moisture content of soils (ASTM, 1985h). (iv) promote microbiological corrosion of buried steel.
	Sulphate content: (a) Total sulphate content of soil (b) Sulphate ion content of groundwater and aqueous soil extracts	These tests assess the aggressiveness of soil and groundwater to buried concrete and steel.
	Total sulphide content of groundwater and soil extracts	Assesses the aggressiveness of soil and groundwater to buried steel.
	pH value	Assesses the aggressiveness of soil and groundwater to buried concrete and steel.
	Chloride ion contents	Assesses: (i) the aggressiveness of soil to buried concrete and steel. (ii) the suitability of fine aggregate for use in concrete.

Table 3.3. (Continued).

Category of Test	Name of Test	Remarks
Soil Strength Tests	(a) Unconfined Compression Test (UCT) (b) Unconsolidated Undrained Triaxial Compression Test (UU) (c) Consolidated Undrained Triaxial Compression Test with Pore Pressure Measurement (CU) (d) Consolidated Drained Triaxial Compression Test with Volume Change Measurement (CD)	Total stress strength parameters of undrained shear strength for cohesive soils can be derived directly or indirectly from laboratory tests. If insufficient undisturbed soil samples are collected, preliminary estimation of shear strength can be derived by correlating to results of Atterberg Limit Tests [42]. Atterberg limits define the boundaries of several states (solid, semi-solid, plastic and liquid) of consistency for plastic soils. The boundaries are defined by the amount of water a soil needs to be at one of those boundaries. For saturated clays with undrained shear strength less than about 75 kPa, the <i>in situ</i> penetration vane test, used in conjunction with the cone penetration test, will normally be the best method for measuring undrained shear strength.
	Direct Shear test (Figure 3.12)	A useful and practical alternative to the consolidated drained triaxial test for shear strength measurements on fill, colluvium and soils derived from weathering of rock <i>in situ</i> . The test specimen can be oriented to measure shear strength on a pre-determined plane.
Soil Deformation Tests	Consolidation (a) One-dimensional consolidation (Oedometer test) (b) Triaxial consolidation (c) Rowe cell	Significant settlements may occur due to consolidation of soil under superimposed load. Consolidation of soft compressible soils involves the removal of excess pore water, thereby reducing the bulk volume. These tests yield soil parameters from which the amount and time scale of settlements can be calculated.
	Modulus of deformation	Values of the modulus of deformation of soil can be obtained from the stress-strain curves from triaxial compression tests, where the test specimens have been consolidated under effective stresses corresponding to those in the field [31].



Figure 3.11. A Laboratory Vane Shear tester (courtesy: ELE International).



Figure 3.12. Shear Strength tester.

surveyor is involved in the works for (b) and (c). The contractor is also required to provide the following:

Accommodation

- first aid
- washing facilities
- meal room
- sanitary facilities

Storage

- cover/no cover
- durability
- security

Fencing

- security

Hoarding

- safety

Electricity & Water Supply

- portable self-powered generator
- metered supply from local authority

Details of these are discussed in Chapter 6, “Materials Handling and Mechanisation”.

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